

Magic Kitchens Meat Loaf Mix

Sales Forecast & Marketing Budget Recommendation

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Agenda

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The Business Challenge

Why Sally needs a model — and why past data is confusing

02

Data & Exploration

24 quarters, 4 variables, hidden patterns

03

Methodology

4 regression models built, 1 selected by Adjusted R^2

04

Key Findings

What actually drives sales — and what doesn't

05

Recommendation

Where to spend \$1K, what to forecast, what to watch

The Business Challenge

What Sally inherited on Day 1

- Meat loaf mix sales swing from \$227K to \$703K per quarter — a nearly 3× gap
- Company policy: EITHER promote OR advertise each quarter (never both)
- Long-standing internal debate: does promotion actually work, or just steal future sales?
- Advertising effect unclear — two similar ad spends (\$36K vs \$39K) produced \$648K and \$343K
- Economist claims product is counter-cyclical — sells better in recessions
- Possible seasonal effect — cold months may drive more purchases

\$227K–\$703K

Sales range

85%

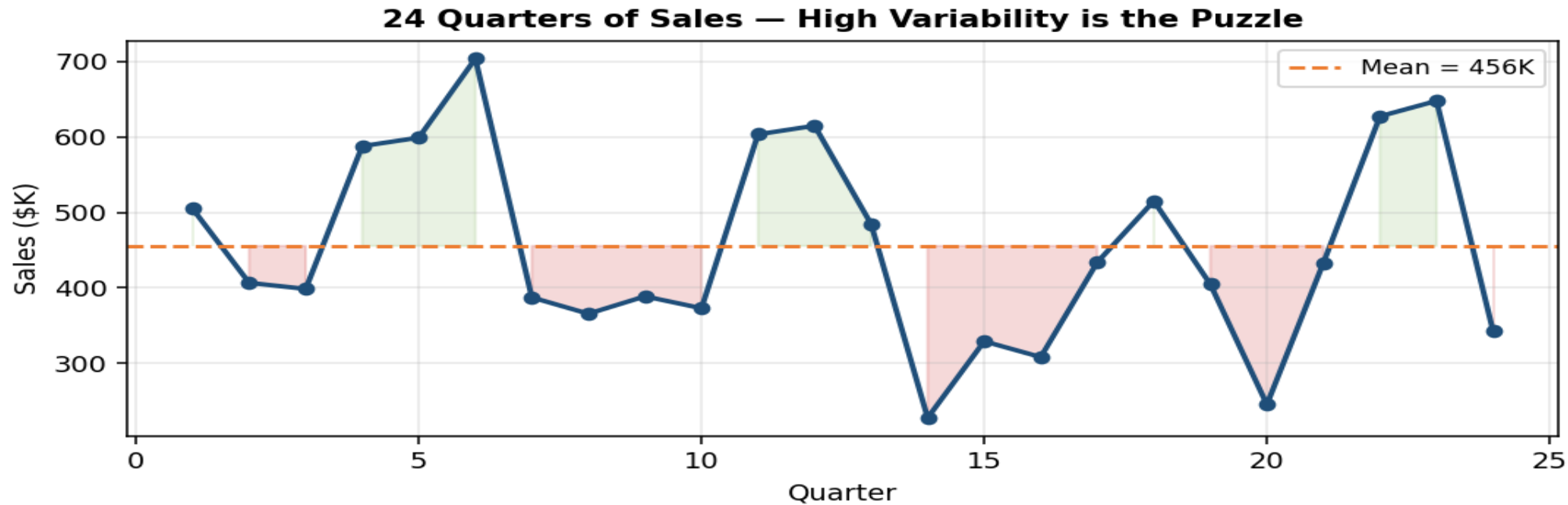
Variance explained
by our model

24 qtrs

Historical data

The Data — 24 Quarters of History

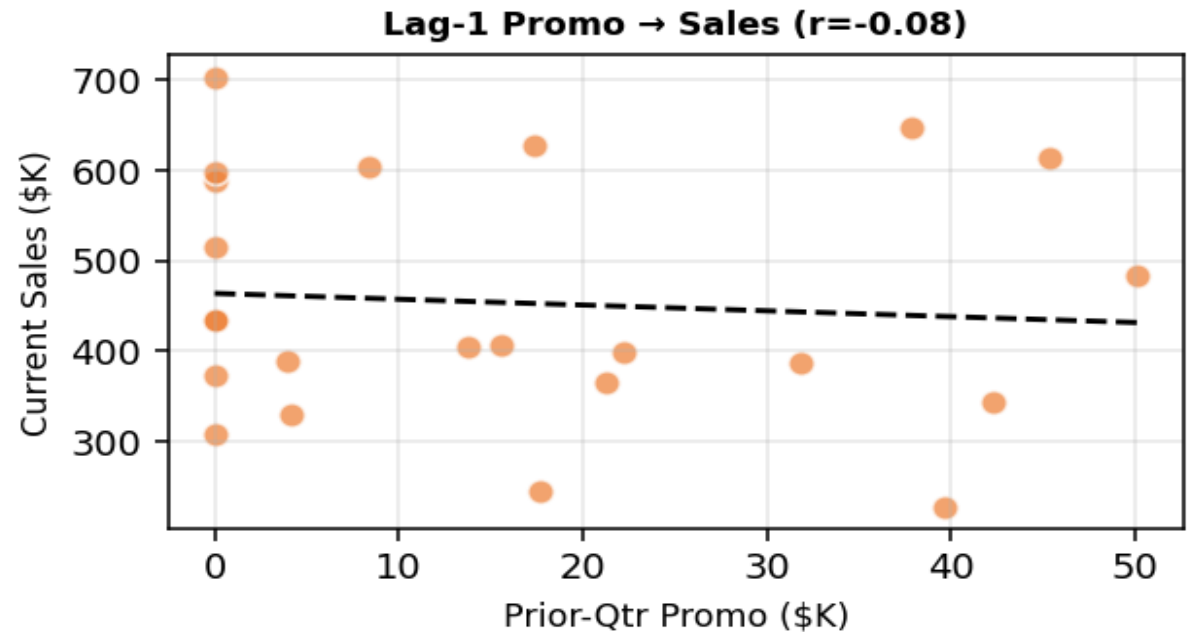
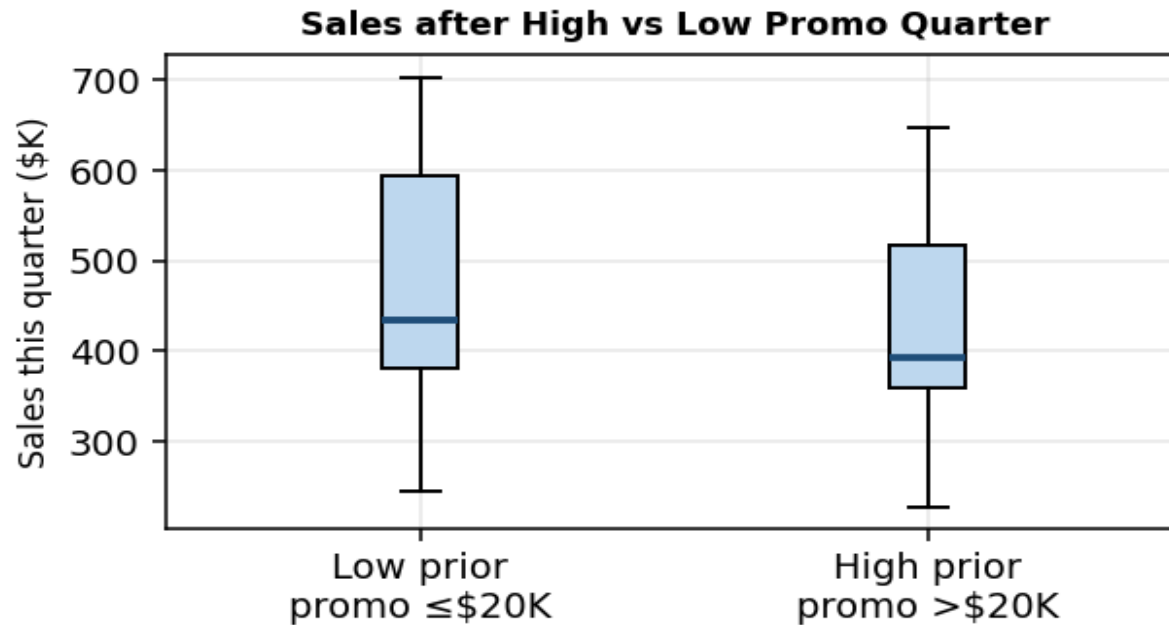
Variable	Description	Mean	Min	Max
sales	Quarterly sales (\$K)	456K	228K	704K
prom	Promotion spend — to brokers (\$K)	15.9K	0K	50.1K
adv	Advertising spend — to consumers (\$K)	26.6K	0K	57K
index	Economic conditions index	105.5	96	114



Key Insight: Promotion Creates a Hangover

What the raw data tells us before we even run a model

- Promotion boosts sales THIS quarter — but suppresses NEXT quarter
- When prior-quarter promo > \$20K → next quarter median sales DROP ~\$80K
- Brokers stockpile inventory during deals, then don't reorder until stock clears
- → This is why we must test lagged (delayed) effects in the model



Model Building — Progressive Complexity

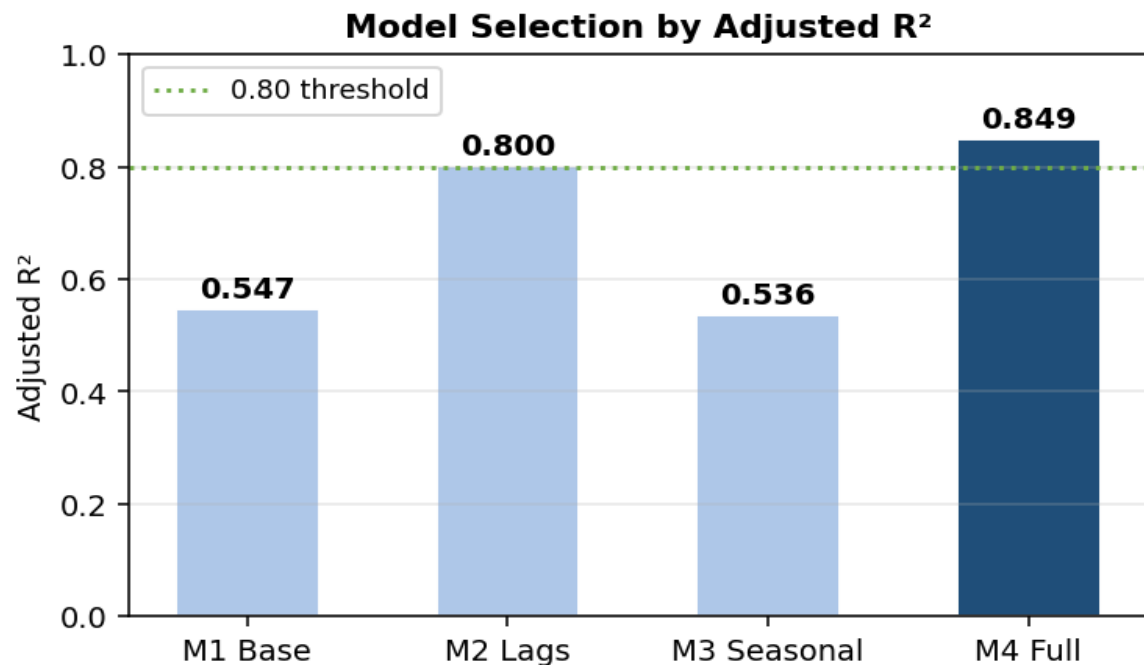
We built 4 OLS regression models, selecting by Adjusted R^2

M1	Base	<i>prom + adv + index</i>	Adj R^2 = 0.547
		✗ Missing lag effects	
M2	Lags	<i>+ prom_lag₁ + adv_lag₁</i>	Adj R^2 = 0.800
		✗ Missing seasons	
M3	Seasonal	<i>+ Q1/Q2/Q3 dummies</i>	Adj R^2 = 0.536
		✗ Missing lags	
M4	Full ★	<i>All of the above</i> ← BEST	Adj R^2 = 0.849
		✓ All effects tested	

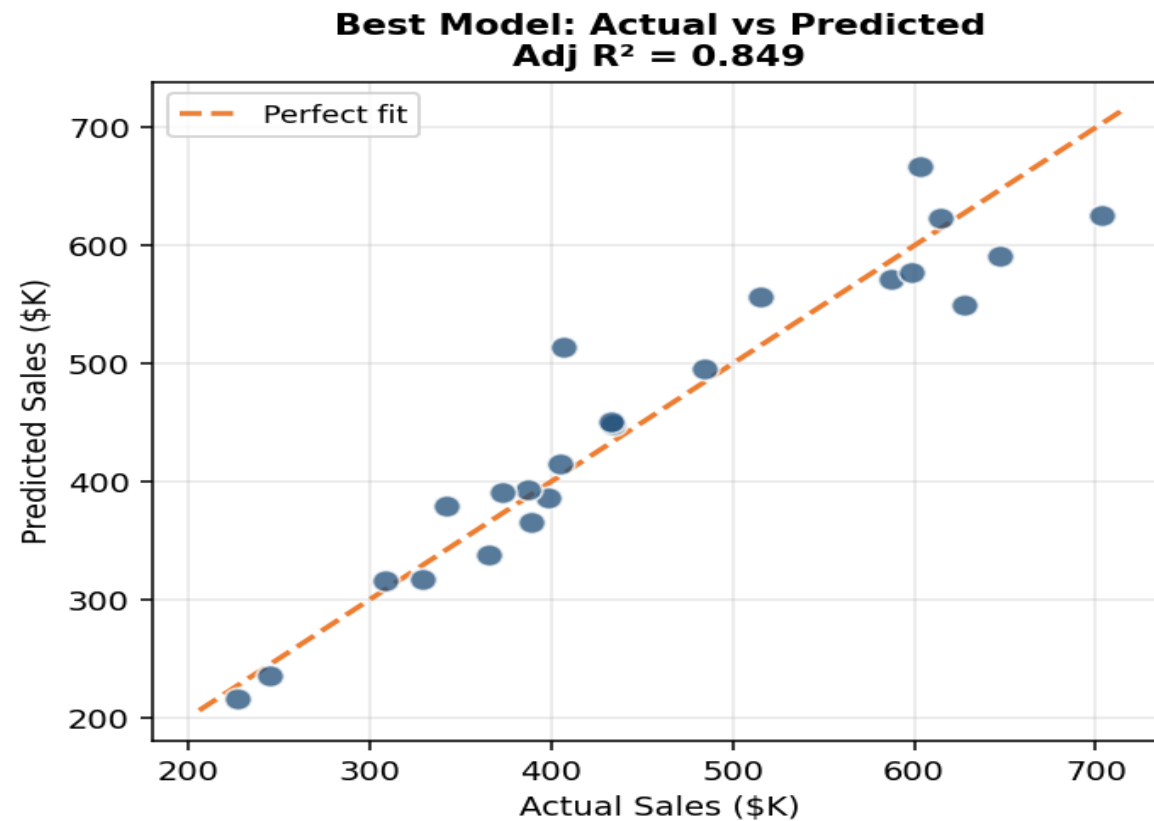
★ Winner: M4 Full — Adj R^2 = 0.850 (explains 85% of sales variance)

Best Model Performance — M4 Full

$Adj R^2 = 0.849 \cdot n = 23 \text{ quarters} \cdot AIC = 254$



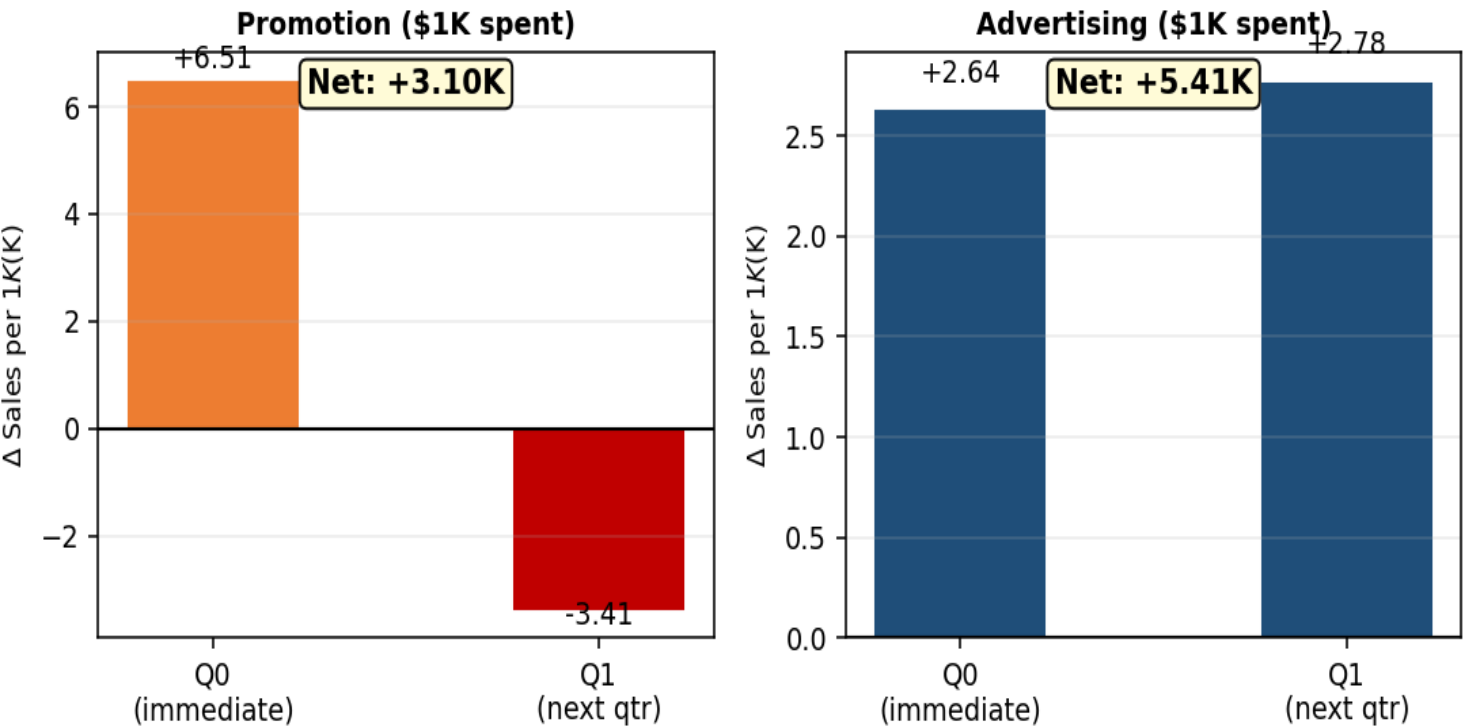
- RMSE: \$40.7K (avg forecast error)
- Residuals are well-behaved — no systematic pattern
- No multicollinearity detected (max $|r| = 0.46$ between predictors)



Q1: Where Should Sally Spend \$1,000?

Answer: Advertising yields a higher net sales return

Two-Quarter Sales Impact per \$1K Spent



Promotion — Net Effect

$+6.51K - 3.41K = +3.10K \text{ net}$

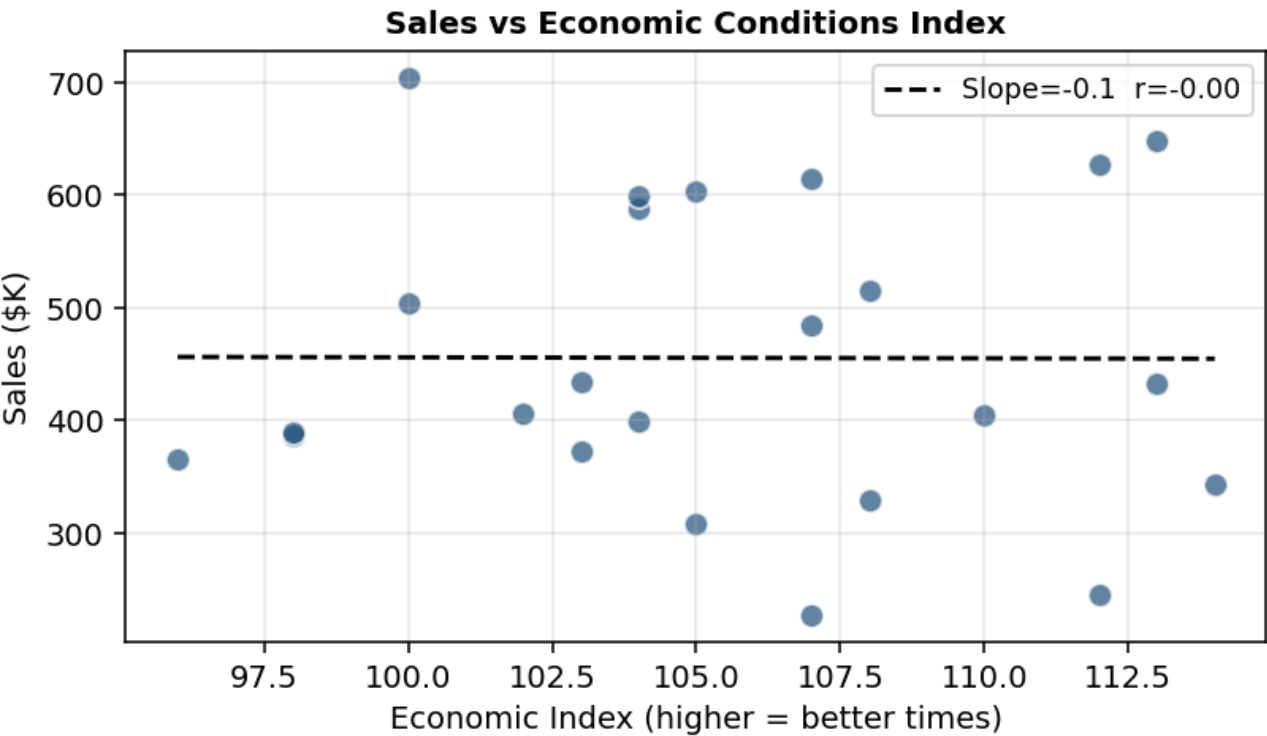
Advertising — Net Effect

$+2.64K + 2.78K = +5.41K \text{ net}$

- Promotion boosts sales NOW (+\$6.51K) but creates a hangover
- Lagged promo is negative (−\$3.41K): brokers stockpile, then go quiet
- Advertising compounds: +\$2.64K now AND +\$2.78K next quarter
- ★ Sally should ADVERTISE — net return is \$5.41K vs \$3.10K

Q2: Is Meat Loaf Counter-Cyclical?

The economist's theory is plausible — but just misses statistical confirmation

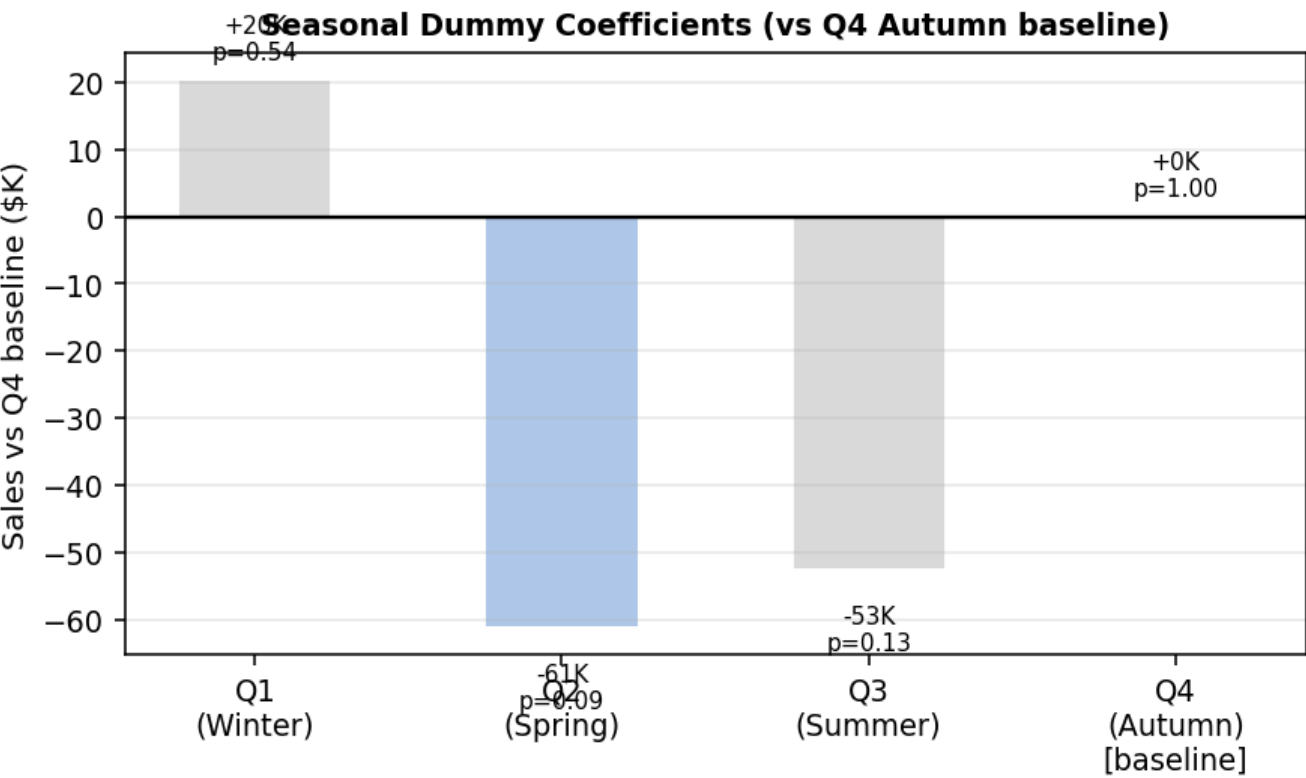


Coefficient (index)	-4.922
t-statistic	-2.11
p-value	0.0530
95% CI	[-9.92, 0.07]

- Coefficient is NEGATIVE (-4.92): ✓ correct direction
- p = 0.053 — misses 5% threshold by a whisker
- Verdict: Suggestive, not conclusive
- Watch economic conditions; cold winters may help

Q3: Are There Seasonal Effects?

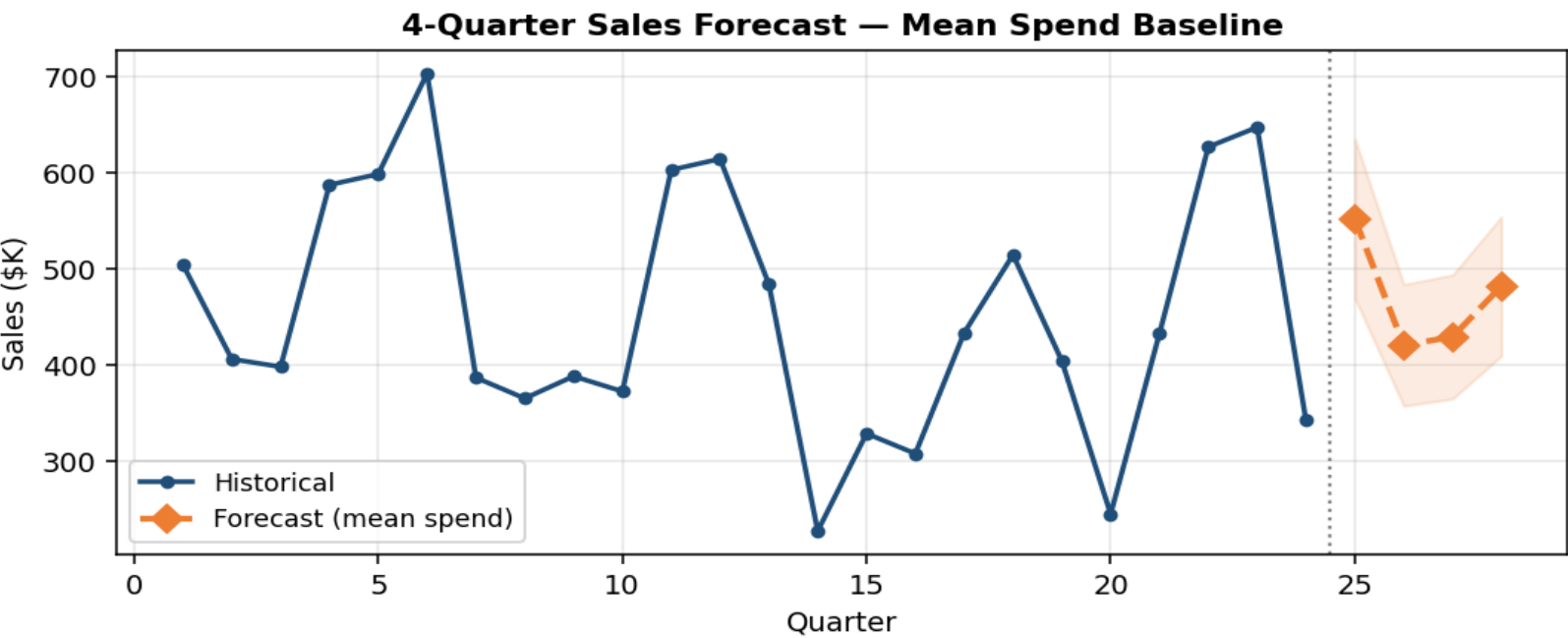
No season significantly outperforms another — after controlling for spend



- Q1 (Winter): +20K vs Q4 — not significant (p=0.54)
- Q2 (Spring): -61K vs Q4 — marginal (p=0.09)
- Q3 (Summer): -53K vs Q4 — not significant (p=0.14)
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- After controlling for spend & economy,
- seasons cannot be confirmed statistically
- (24 quarters may be too few to isolate)
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- → Do NOT plan seasonal budget shifts yet
- → Re-evaluate with more data

4-Quarter Sales Forecast

Based on M4 Full model — historical mean spend as baseline



- Annual forecast: \$1885K (mean-spend baseline)
- To maximise: shift spend toward advertising
- Monitor economic index — weak economy may boost sales

Quarter	Forecast (\$K)
Q1 (Year 7)	\$552.5K
Q2 (Year 7)	\$421.0K
Q3 (Year 7)	\$429.6K
Q4 (Year 7)	\$482.3K
Annual Total	\$1885.5K

Sally's Action Plan

Three evidence-based decisions for the coming year

Shift budget toward Advertising

- Advertising net effect: +\$5.41K per \$1K — nearly 2× promotion
- Eliminate or reduce promotion-heavy quarters
- Use promotion sparingly (max 1–2 quarters) to avoid stockpiling hangover

Monitor economic conditions

- Index coefficient is negative: bad times → more meat loaf sold
- Keep a recession-ready plan; this product may be a safe harbour
- Re-test counter-cyclicalities as more data accumulates

Do not yet plan by season

- Seasonal patterns are unconfirmed at 5% significance
- Collect 2 more years of data before allocating seasonal budgets
- Track Q2 (spring) — shows the largest (if marginal) negative effect

OLS Regression Output — Best Model: M4 Full (Adj R² = 0.850)

Dependent Variable Sales (\$K)	Observations 23	R-squared 0.9042	Adjusted R-squared 0.8495	F-stat p-value 0.0000	AIC 253.72	RMSE \$40.7K
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Variable	Coefficient	Std Error	t-Stat	P-value	[95% CI Low	95% CI High]	Significant?
const	808.2760	242.3750	3.335	0.0049	288.4335	1328.1186	✓ Yes
prom	6.5088	0.8062	8.074	0.0000	4.7798	8.2379	✓ Yes
adv	2.6374	0.7269	3.628	0.0027	1.0784	4.1965	✓ Yes
index	-4.9225	2.3292	-2.113	0.0530	-9.9180	0.0731	No
prom_lag1	-3.4099	0.7851	-4.343	0.0007	-5.0938	-1.7260	✓ Yes
adv_lag1	2.7752	0.7493	3.704	0.0024	1.1682	4.3823	✓ Yes

Q1	20.3601	32.8144	0.620	0.5449	-50.0198	90.7400	No
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Q2	-61.2449	33.7064	-1.817	0.0907	-133.5380	11.0482	No
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Q3	-52.6502	33.1690	-1.587	0.1348	-123.7906	18.4902	No
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Note: Green = statistically significant (p < 0.05). Lagged variables use n=23 (one observation lost to lag).